

Homework 1

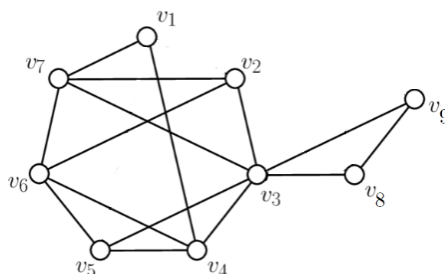
Instructor: Prof. Warnke

Due: Monday 04.06.2024, 8:00am

Please upload your solutions to Gradescope (all graphs are assumed to be simple: i.e., no multiple edges, loops or weights; and only undirected edges). The solution of every problem-part should not be longer than one page. Furthermore, unless specifically told otherwise, you should always prove your assertions; sometimes, this may take the form of a list or a computation, but some form of justification will always be required. If submitting hand-written solutions, write all solutions clearly and very neatly; else, type them up using LaTeX or a similar typesetting package (you cannot get credit for solutions your graders cannot read).

Problem 1

Recall that two distinct vertices u and v are called neighbors if they are the endpoints of an edge; we denote by $N(u)$ the set of all neighbor-vertices of u (see also Section 1.4 in the Verstraete lecture notes, or Section 1.3 in the West textbook). For the graph G depicted below, write down the number of vertices $v(G)$, the number of edges $e(G)$, the neighborhoods $N(v_1)$ and $N(v_5)$, the degrees $d(v_1)$ and $d(v_5)$, and the minimum and maximum degrees $\delta(G)$ and $\Delta(G)$. No need to justify your answers.

**Problem 2**

Find and explain the error in the following (wrong) proof by induction:

(False) Claim: Assume that every vertex of G has at least two neighbors. Then G has at least $2v(G) - 3$ edges.

(Incorrect) Proof: by induction on the number $n = v(G)$ of vertices. For the base case $n = 3$, note that any graph with 3 vertices needs to contain all 3 possible edges to satisfy the assumptions, so G has (at least) $3 = 2 \cdot 3 - 3$ edges. For the induction step $n \geq 4$, assume the claim holds for all graphs with $n - 1$ vertices: if they satisfy the assumption, they all have at least $2(n - 1) - 3 = 2n - 5$ edges. To get a graph G with n vertices, we add a vertex; to make sure that it is the endpoint of at least two different edges, we need to add at least two new edges. Therefore the n -vertex graph G has at least $e(G) \geq 2n - 5 + 2 = 2n - 3$ edges.

Problem 3

(a) Using induction, prove that every n -vertex graph G has at most $e(G) \leq \binom{n}{2}$ edges.

(b) Using induction, prove that for every graph G there exists a partition $V(G) = V_1 \cup V_2$ of its vertex-set so that at least $e(G)/2$ edges of G go between V_1 and V_2 , i.e., have one endpoint in V_1 and one endpoint in V_2 .

(c) A graph H is called a subgraph of a graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$; in this case we write $H \subseteq G$. Using induction, prove that every graph G with $v(G) \geq 7$ vertices and at least $e(G) \geq 5v(G) - 14$ edges contains a subgraph $H \subseteq G$ with minimum degree at least $\delta(H) \geq 6$.

Problem 4

The Handshaking lemma states that $2e(G) = \sum_{v \in V(G)} d_G(v)$. Prove this using induction in two ways:

- (a) by induction on the number of edges $e(G)$, and
- (b) by induction on the number of vertices $v(G)$.

Submission Instructions

1. To submit your completed work, log in to Gradescope with your `@ucsd.edu` email (either directly or through Canvas).
2. Here is a guide to submitting an assignment in Gradescope: <http://bit.ly/3LRFSil>.
3. Please solve each problem on a separate page. If submitting as a single PDF, **be sure to “assign pages”** in your submission to the corresponding problems in Gradescope (otherwise the grader may have trouble finding a problem, and you could lose credit for it).
4. Don't forget to double-check that your work was successfully submitted – **it will take some time to upload your work, so start early!** If you submit work early, you can always update your submission before the deadline.

Induction

Since this is the first homework, here are some review resources on Induction, which you may find useful:

- Discussion 1 sheet contains an ‘induction template’ (available on Canvas)
- Appendix Theorem A.25 and Example A.28 in the Graph Theory textbook by Douglas West
- Appendix A.6 in the lecture notes by Jacque Verstraete (available on Canvas)
- These notes by Matt DeVos